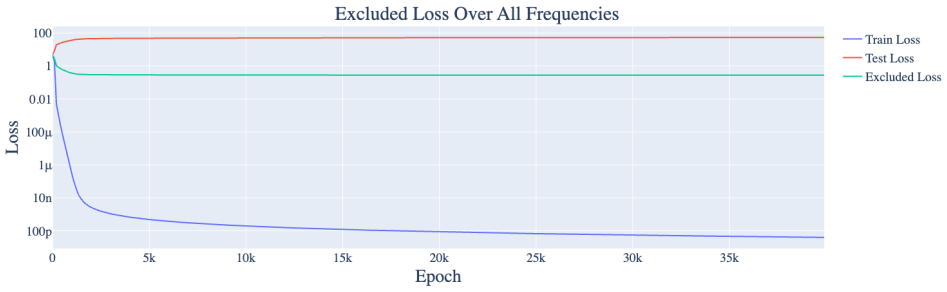

Figure 26: Training, test, and full excluded loss for a 1-layer version of the original architecture without weight decay. One random seed chosen for readability. Note that not having weight decay prevents grokking.

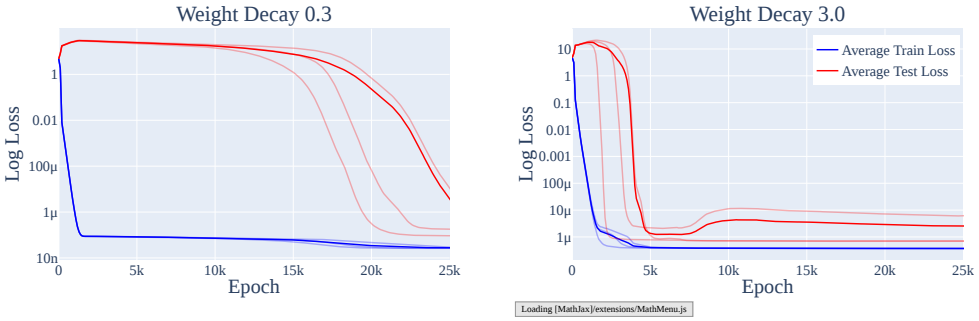

Figure 27: The train and test loss over the course of training with weight decay $\lambda = 0.3$ (left) and $\lambda = 3.0$ (right). Less aggressive weight decay leads to slower grokking.

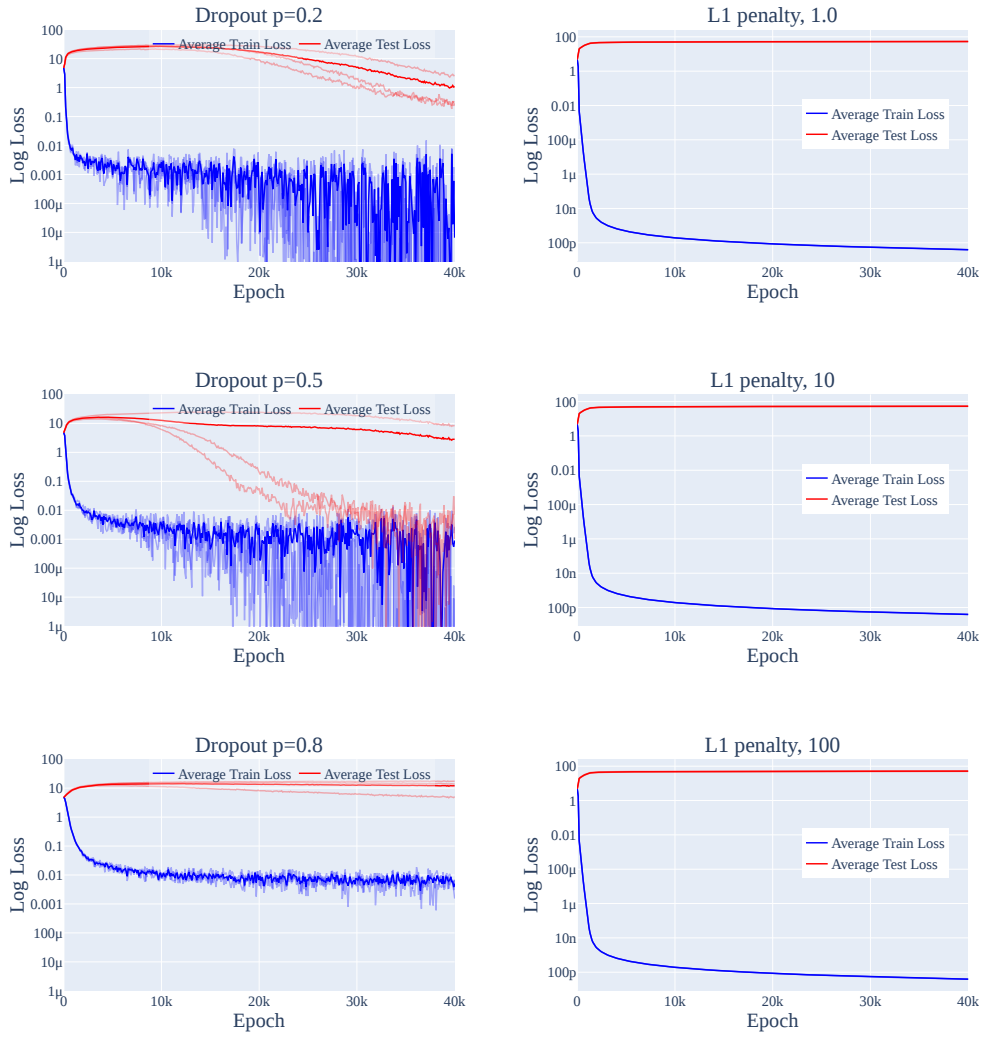

Figure 28: The train and test loss over the course of training with two types of regularization, dropout and ℓ_1 regularization. Grokking occurs with some runs for dropout but never for ℓ_1 regularization.

2. ℓ_1 **Regularization** We add an ℓ_1 penalty to the loss term. We use $\lambda \in \{1, 10, 100\}$. Note that we *do not decouple* the updates with respect to the ℓ_1 penalty from optimization steps done with respect to the log loss (as is done for ℓ_2 regularization via AdamW Loshchilov & Hutter (2017)).

In each case, we ran three random seeds. We show the results in Figure 28. While grokking did not occur with ℓ_1 regularization, we found that it does occur for all three seeds using dropout with $p = 0.2$ or $p = 0.5$. We speculate that this is because both dropout and weight decay encourage the network to spread out computation (which is required for the Fourier multiplication algorithm), while ℓ_1 regularization encourages the network to become more sparse in the neuron basis and thus *less* sparse in the Fourier basis, preventing the network from learning the Fourier Multiplication Algorithm.

D.2 The slingshot mechanism often occurs, but is unnecessary for grokking

As noted in Section C.2, our 2-layer transformers exhibit significant slingshots (Thilak et al., 2022) during training. We speculate that this is due to how gradients of different scale interact with adaptive

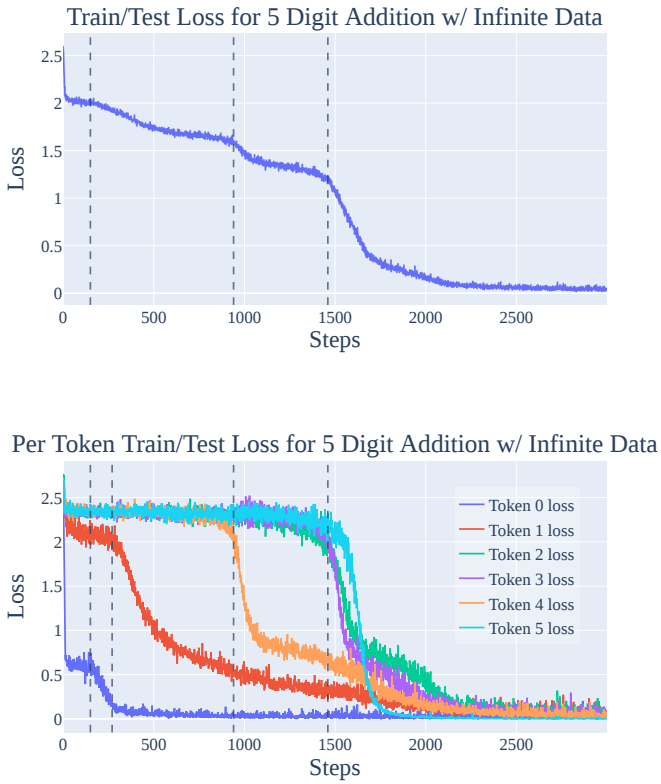

Figure 29: (*Top*) The training/test loss for 5 Digit Addition trained on randomly generated data. Note that training and test loss coincide, as the model does not see repeated pairs.(*Bottom*) The train/test loss *per token* for 5 Digit Addition, trained with randomly generated data at each step. Note that phase changes in the average loss correspond to phase changes in individual tokens, though one phase change (token 1, around step 270) is not visible on the averaged loss as it overlaps with the end of the first phase change (token 0, starting around step 150).

optimizers. We were even able to induce slingshots on a 1-layer by reducing the precision of the loss calculations (as this causes many gradients to round to 0 and thus greatly increases the differences in scale of gradients).

However, as many of our 1-layer models do not exhibit slingshots but nonetheless grok, the slingshot mechanism is unnecessary for grokking to occur, in the presence of weight decay or other regularization. We speculate that the slingshots of Thilak et al. (2022) (which co-occur with grokking for training runs *without* weight decay) serve as an implicit regularization mechanism that favors the simpler, generalizing solution over the more complicated

D.3 Additional evidence from other algorithmic tasks

We now provide addition analysis of grokking phenomena on 3 additional algorithmic tasks and confirm that limited data is an important part of grokking:

1. **5 digit addition.** We sample pairs of random 5 digit numbers and have the model predict their sum
2. **Predicting repeated subsequences.** We take a uniform random sequence of tokens, randomly choose a subsequence to repeat, and train the model to predict the repeated tokens.
3. **Skip trigram.** We feed in a sequence of tokens from 0 to 19, of which exactly one is greater than or equal to 10, and the model needs to output the token that is ≥ 10. This can be solved with learning 10 skip trigrams.